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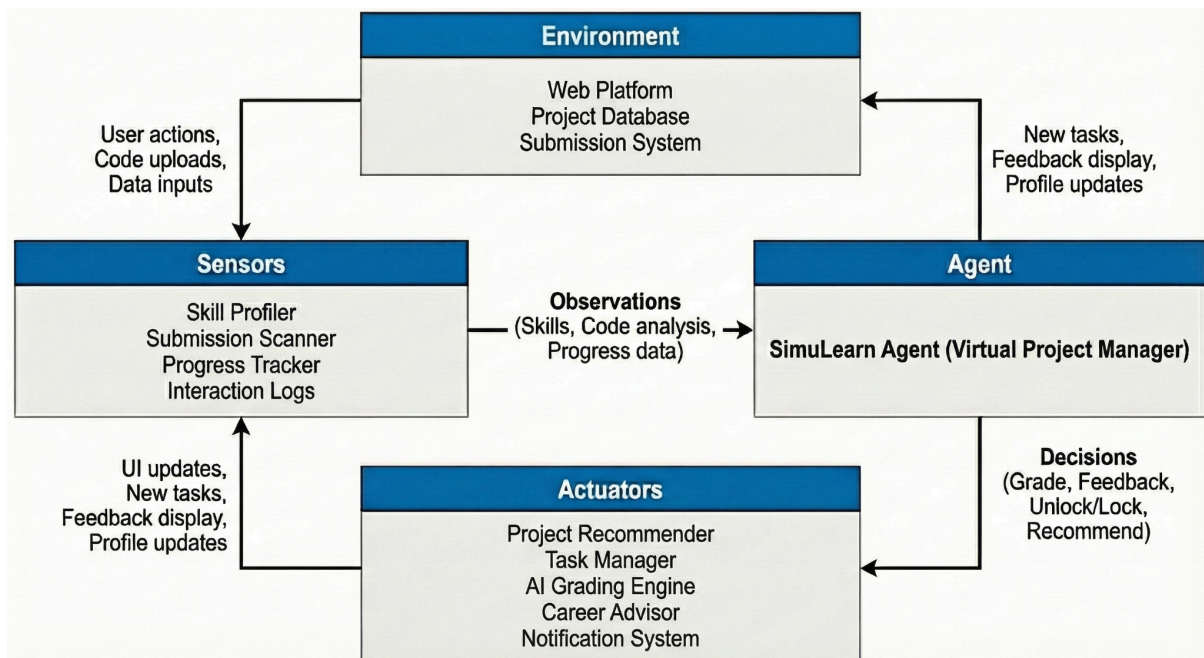
1.0 PEAS Model

SimuLearn acts as an intelligent Virtual Sensor Project Manager, designed to guide user through realistic industry simulations.

Property	Description
Performance	● Acquire User Skills: Ensure users learn new technical skills. ● Grade Accurately: Evaluate code when compared against predefined industry standards. ● Recommend Relevantly: Suggest projects that match the user's likes and needs. ● Provide Quality Feedback: Provide clear, and helpful advice after task or project submission. ● Engage Users: Task and project completion rates, indicating sustained learner motivation.
Environment	● Web Platform: The SimuLearn dashboard, project library, skill profile, and task workflow interface. ● Project Database: Repository of simulated real-world projects, task dependencies, constraints, and evaluation criteria. ● Submission System: Interface that receives source code files, textual explanations, and user inputs for assessment.
Actuators	● Project Recommender: Suggests suitable projects based on inferred user skills and learning gaps. ● Task Manager: Unlocks or blocks task nodes according to prerequisite satisfaction and learning progression rules. ● AI Grading Engine: Automatically evaluates submissions, assigns scores, and generates structured feedback. ● Career Advisor: Updates user skill profiles and provides personalized career path suggestions. ● Notification System: Sends system messages such as pass or fail alerts.
Sensors	● Skill Profiler: Collects self-reported skills, syllabus topics, and free-text input from users.

	• Submission Scanner: Analyzes uploaded code structure, syntax, and compliance with task requirements. • Progress Tracker: Monitors task completion status, time-on-task, and retries. • Interaction Logs: Tracks user interactions such as hints accessed, and resources viewed.
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2.0 Relations Between Each Property



The diagram shows how each property interacts with each other.

1. **Environment -> Sensors:** The cycle begins in the **Environment**, which is the SimuLearn Web Platform when a user interacts with the system. This includes the actions like uploading solutions, entering skill data, or navigating the dashboard.
2. **Sensors -> Agent:** The **Sensors** detect the user actions. Then it will process raw inputs like a code file and text and turn them into "**Observations**", which include specific data like user's skills, code solution, and progress. Then these data will be

sent to the Agent.

3. **Agent -> Actuators:** The **Agent** will receive these observations. Then it uses its internal logic and rules to process the data and make "**Decisions**". These decisions determine what needs to happen next, like grading a submission, generating specific feedback, and deciding to unlock the next task.
4. **Actuators -> Environment:** The Agent sends commands based on its decisions to the **Actuators**. The actuators execute these commands and change the state in the **Environment**. For example, it will update the UI with the grade, display feedback and enable a new task button. This updated environment sets the stage for the user's next interaction then restart the cycle.

3.0 Explanation In Proof Of Concept (POC)

Performance: In Simulearn, performance is measured by the result of the AI Grading Engine. When a user submits the solution, the system will analyse and check it with a strict checklist such as code security, functionality and best practices. Then, the system calculates a score based on these checks. If the user consistently passes tasks that are rated as "Hard", the system ensures that he is acquiring new skills, proving the agent is performing its role effectively in guiding the user.

Environment: The environment includes static data and user interactions. The Project Database consists of all simulation scenarios, while the Web Platform is dynamic, changes based on user inputs. The system also acknowledges code files from the user, making the simulation accessible and realistic. The architecture is built on using a strict logical structure that ensures users follow the correct order of tasks and avoids them from skipping important steps.

Actuators: The actuators are implemented as services within the Backend System that change the interface. The Task Manager actuator modifies the dashboard button from the gray "Locked" to the green "Open". Besides, the AI Grading Engine actuator will generate

text-based feedback reports to users. In our demo, these actuators react quickly to user actions based on the agent's decisions.

Sensors: The sensors are the entry point of how the system collects and retrieves the information. The Submission Scanner reads the files a user uploads to check whether the correct files are present. The Skill Profiler reads the text of user's inputs to identify their confidence levels. These sensors turn the user's actions into data that the AI can understand and process into a result.

4.0 Behaviors Of Agent To Achieve The System Goals

To fulfill the objective of bridging the gap between academic theory and real-world industry practice, the SimuLearn agent operated through four different roles:

1. Interviewer (Onboarding & Matching)

At the onboarding stage, the agent functions as an interviewer. Using the Skill Profiler sensor, it evaluates the user's existing knowledge and competencies. Rather than assigning projects randomly, the agent computes a match score for each available scenario. This ensures that user are assigned tasks that are appropriately challenging and difficult enough but not so hard to promote learning without frustration.

2. Gatekeeper (Workflow Management)

During the project, the agent acts like a strict gatekeeper. It monitors the project's task list. Even if a user feels ready for advanced tasks, the agents keep them locked until the necessary previous tasks are done. This mimics a real-world manager who ensures work is done in the right order, teaching the user the importance of following a proper process and correct task order.

3. Code Reviewer (Evaluation)

When a user submits their solution, the agent acts as a code reviewer. It analyzed the submission by checking for specific keywords, common errors and good coding practices, then compared it against an expected solution. If the evaluation score falls

below 60%, the submission is rejected. This rejection is intentional and designed to encourage iteration and improvement which represents a professional software development environment.

4. Mentor (Feedback & Growth)

Once a task is successfully completed, the agent assumes the role of a mentor. Its main goal is to review the user's solution and give helpful feedback to support their learning. Moreover, the agent looks at the user's solution and suggests career paths that match their real strengths.